Beyond the Naked Eye: Automated Delivery Analysis in Cricket with Computer Vision

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***Abstract*—Computer vision has really changed how we track and analyze performance in sports. By using automated imaging feedback, coaches can assess players, spot key events, and rethink strategies far faster than by watching tape alone. Cricket provides a vivid case study: researchers now employ neural networks for ball tracking, pose estimation, and play identification. Yet most commercial setups cost a small fortune, because they bundle multiple synchronized cameras with specialist hardware that grassroots clubs cannot afford. To bridge that gap, the present work describes a single-camera, budget-minded system that still delivers detailed ball-tracking data. A bespoke computer-vision pipeline stitches together off-the-shelf components and open-source code. Core routines include a tuned YOLOv5 detector, a linear Kalman filter, and a custom homography stage that maps field coordinates from standard video. Rigorous backyard trials confirm the pipeline dishes out useful objective metrics, but uneven homography calibration remains the fly in the ointment as it drags accuracy down when the camera angle shifts quickly. The system proves itself as a lightweight coaching aide, not as a rival to the super-slo-mo rigs seen on television. Moving forward, we'll need to tackle these issues like bouncing camera mounts, glare, and shifting light conditions while also expanding the analysis tools so community clubs can gain deeper insights without stretching their budgets.**

***Index Terms*—Computer Vision, Deep Learning, Cricket Analytics, Object Tracking, YOLOv5, Kalman Filter, Sports Technology, Homography.**

I. Introduction

Data analysis has revolutionized elite-level cricket through its application of computer vision systems which provide exact data to replace subjective decision-making. The modern technology has developed ball tracking, automated umpiring, and player pose estimation systems which generate detailed performance analysis [2], [5], [7]. The Hawk-Eye system and other top analytical systems require precise data input which needs multiple synchronized cameras and high-performance processing units that work independently [9], [12]. The system creates two main barriers which prevent entry into the market. This is because of its high costs and complex technical requirements. The lack of objective performance data prevents grassroots clubs and community teams, and coaches with limited budgets from accessing reliable information, so they must depend on slow and subjective video analysis methods which prove unreliable [4], [13].

The major difference between these systems drives the entire goal of this research because it needs to create a budget-friendly single-camera solution which provides instant ball tracking data. Our primary objective involves developing a system that generates exact pitch location data and delivery type results through a single-camera system which operates independently from synchronized multi-camera systems and expensive custom hardware. The system functions as a lightweight user-friendly coaching assistant tool because it operates as a lightweight user-friendly coaching assistant tool. The goal is to provide professional sports analytics data to amateur athletes.

The system functions through a single computer vision pipeline which combines open-source elements with commercial components. The system operates by uniting three essential technical operations into a single integrated framework. The system achieves real-time ball localization through its use of a specially optimized *YOLOv5s detector* which operates effectively on mid-range computing hardware. The system uses a *linear Kalman filter* to smooth the trajectory estimation which helps maintain track continuity during brief occlusions or high motion blur. The third component operates as the most fundamental element for single-camera metric

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analysis because it implements a customized *homography transformation* process which converts 2D pixel coordinates into exact 3D field measurements (meters on the pitch).

The primary contributions of this work are four-fold:

1) The development and implementation of an integrated open-source pipeline that achieves robust ball-tracking and delivery classification using a single, fixed camera, thus overcoming the cost and mechanical complications associated with synchronizing multiple cameras.
2) A technical justification of the two-step Kalman Filter cycle and the YOLOv5s detector is presented with quantitative analysis of runtime performance versus alternative tracking and detection methods.
3) The presentation of a detailed analysis, quantifying the error introduced by the homography projection, which defines the accuracy limits of the system in spatial analysis.
4) Delivery of a low-latency, user-friendly tool that converts raw video into objective, decimal-based coaching metrics (e.g., delivery length classification) digestible to players minutes after a session is finished.

The rest of the paper describes the methodology and presents extensive experimental results with necessary benchmarking. It also discusses the system's limitations and future extensions.

II. Related Work

Automated officiating, individual performance analysis, and strategic scene interpretation are the three main categories of computer vision-based cricket analytics research. Our contribution is anchored by prior work that emphasizes the considerable cost and complexity that are frequently involved in obtaining the necessary metric precision.

A. Automated Umpiring and Event Detection

Research on automated officiating, such as Crick-Net [12] for no-ball spotting and the comprehensive third-umpire package by Iyer and team [7], endeavors to resolve close calls with high precision. The research continues to progress toward the development of automated systems which perform umpire signal transcription and event logging through convolutional architectural methods [2], [3]. These systems operate successfully yet they need specific settings to function which includes high-speed cameras and multiple synchronized inputs for precise sub-millimeter accuracy. The main barrier to grassroots implementation emerges from the requirement to use expensive advanced equipment for precise metric computation which our single-camera solution aims to eliminate.

B. Player and Performance Analysis

A similar line of investigation incorporates computer vision to examine the exact mechanics of individual players. Researchers combine human-pose models with cricket video to highlight posture and motion, making empirical data available to coaches on bowling and batting forms [10], [13]. Some efforts have managed to capture the exact backlift phase of a bat's swing [4], [8] while other tools classify bowler types and extract action data sets at scale [5], [6]. These analyses often require either specialized 3D pose tracking equipment or extensive labeled data sets that are unavailable to community clubs [15], even while highlighting the value of granular data.

C. Ball and Scene-Level Analysis for Strategic Insights

Ball tracking functions as the core element for strategic analysis. The deep-learning pipeline created by Abbas and his team demonstrated that imperfect cameras could identify deliveries in broadcast footage before reconstructing their movement through frame-by-frame analysis [1]. The research of Moodley and Suguna applied RetinaNet architectures to construct complete scene maps for strategic prediction purposes [9], [14]. The methods provide excellent scene awareness but they handle ball tracking through 2D methods which fail to apply homography transformation for accurate pitch coordinate mapping required by metric delivery classification [11]. Our research develops automated solutions to convert 2D data into 3D metrics through a cost-effective single-camera system.

III. Proposed Methodology

Central to this project is a single data pipeline, developed to scan the raw footage of a single cricket delivery and make sense of chaotic pixel streams by producing digestible, metric-based insights. The approach unfolds in an orderly sequence of steps from chaotic pixel streams to digestible, location-specific insights. The flow of the entire system architecture is presented in Fig. 1. Every stage of this pipeline is discussed in subsequent subsections.

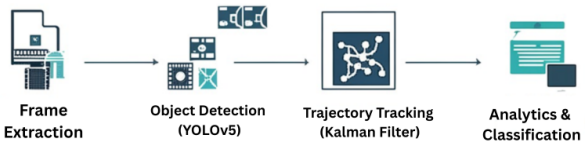


Fig. 1: The overall system architecture, illustrating the flow from raw video input through detection, tracking, mapping, and final analysis.

A. Frame Extraction and Pre-processing

The pipeline uses the input standard digital video, for example, .mp4, from a single fixed camera. OpenCV is used to slice the video into individual RGB frames for sequential processing. With the aim of near real-time performance, pre-processing routines like white balancing and histogram fixes are intentionally omitted here, deferring to robustness in the downstream detection model.

B. Ball Detection using YOLOv5

Locating the cricket ball within a video frame remains the linchpin of the tracking problem. To that end, we reach for YOLOv5, a single-shot detection model that delivers speed without sacrificing too much precision. More precisely, the work here leans on the yolov5s variant-which is small enough to run on a mid-range laptop and already weighted on the COCO suite. The pre-training process contains the coarse class "sports ball" which eliminates our need to perform new annotation and optimization work.

The software performs one forward pass for every new image which produces the coordinates of detected targets. Each detection consists of a bounding box, defined by its top-left corner coordinates (x, y) and its width and height (w, h), along with a class label and a confidence score. Only detections scoring above 0.3 slip past the first filter, a ceiling tight enough to trim most background noise yet lenient enough to catch borderline hits. Should two or more candidates occupy the same pixel patch the algorithm keeps the one boasting the highest certainty, under the working assumption that replica blobs seldom survive unspoiled camera optics. Section V-A gives details on the quantitative comparisons with regards to runtime (FPS) and precision (mAP) of the YOLOv5s variant against alternative single-shot detectors.

C. Trajectory Smoothing with a Kalman Filter

YOLOv5 does a great job of finding objects in space, but regular video can have any number of problems: there can be motion blur, or that moment when a player blocks the ball from view. To create a believable, smooth route for the ball, we use a standard linear Kalman filter to lessen jitters. The filter maintains a state vector for the ball, which incorporates its 2D position (p_x, p_y) and its velocity (v_x, v_y). For every frame, the filter runs in a continuous two-step cycle:

1) **Prediction Step:** The filter projects where it thinks the ball would land today from the last recorded state to fill the void created due to any missed detections.
2) **Update Step:** If YOLOv5 detects the ball, then the centers of that bounding box provide a new measurement. The filter unites the estimate with the new data during this update stage to adjust its internal parameters which results in a trajectory that reflects the most current precise information.

The two-step process forms the foundation which supports both the durability and operational stability of the Kalman Filter. The Prediction Step operates as a vital component for maintaining tracking when objects become hidden and for controlling motion blur effects but the update step performs immediate state modifications by using recent detection information. The runtime analysis in Section V-B shows that this strategy is computationally lightweight and necessary to generate a smooth, credible trajectory in noisy video circumstances.

D. Real-World Mapping via Homography

The single-camera system achieves metric precision through homography transformation which has been used in multi-camera triangulation systems for this purpose. The process of mapping 2D pixel coordinates onto a predetermined 2D pitch plane enables us to avoid using stereo vision systems that require both complex calculations and mechanical synchronization to determine depth information. The approach reduces hardware needs but maintains sufficient spatial precision to conduct performance evaluations.

Of particular interest is the capability to transform the flight of a cricket ball in any given video frame into accurate 3D coordinates on the pitch. Using a homography transformation, as shown in Fig. 2, this can be done once a given setup of the camera is calibrated.

The operator selects, say, four easily identifiable points on the playing surface-the corners of the batting crease-then logs their pixel positions and corresponding real-world coordinates in meters. We compute a 3×3 perspective transformation matrix, $\mathbf{H}$, given the pixel coordinates of these points and their corresponding physical coordinates in meters. This matrix encapsulates the camera's perspective-its position, orientation, and focal length. Once computed, this matrix $\mathbf{H}$ is used to project every tracked 2D point (u, v) from the ball's trajectory onto a top-down 2D plane representing the pitch, yielding its real-world coordinates (X, Y). The projection step is necessary for all subsequent spatial analyses the study intends to perform.

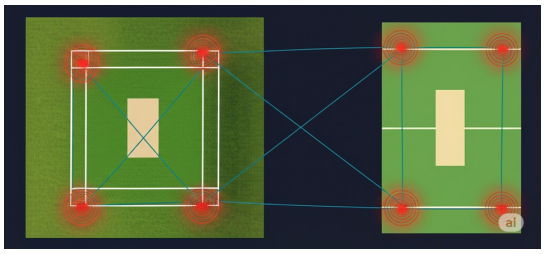


Fig. 2: An example of the homography transformation conceptually. A 2D top-down depiction of the cricket pitch is created by mapping points from the 2D camera view to their respective locations.

IV. Dataset and Implementation Details

A. Dataset and Pre-processing

Publicly available videos of cricket matches were the material for our qualitative evaluation and system analysis. For the main data, we have extracted standard-definition (SD) video clips from a television broadcast stream using a side view camera locked during ball delivery. This approach is in contrast to high-quality, purpose-shot videos that test the real-world robustness of the system. The number of samples in the dataset is bound for large-scale generalization, though it was intentionally prepared to be sufficient for validation of the

feasibility of the pipeline and its tracking accuracy in real-time for heterogeneous low-quality inputs.

This is dependent on a fixed camera angle, providing a strict frame of reference and allowing the precomputed homography from first to last frame to remain valid. The images have not been equalized in color space or normalized in any way during this trial. YOLOv5 was tuned for the type of lighting and picture quality commonly found in day-to-day broadcasts. In the interest of full disclosure and to provide the reader with an opportunity for reproducibility, specific video clips used for the quantitative analysis were obtained from the public repository https://www.kaggle.com/datasets/shujaanazhar/cricket-videos-of-legal-wide-lbw-and-no-balls.

B. Implementation Environment and Framework

The complete software stack ran on Python 3.10 within a virtual environment. Heavy lifting for neural tasks rested on the PyTorch framework, providing fast tensor computation for the YOLOv5 model. The complete range of image and video processing operations including file I/O and frame extraction and annotated video rendering were performed using the powerful OpenCV library. The computationally intensive tasks of matrix operations involving the Kalman Filter and homography transformation were done with specialized functions of NumPy. We used Pandas to structure and sequentially log the analytical output into a persistent CSV file. The key technologies are summarized in Table I.

TABLE I: Core System Components and Technologies

Component	Technology/Library	Role in System
Detection	YOLOv5s (PyTorch)	Performs real-time detection of the cricket ball in video frames.
Tracking	Kalman Filter (NumPy)	Manages state estimation and provides a smoothed trajectory of the ball.
Video I/O	OpenCV	Handles reading of video files, frame-by-frame processing, and writing the final annotated output video.
Mapping	Homography (OpenCV)	Computes and applies the perspective transformation from 2D pixel space to 3D world coordinates.
Data Logging	Pandas	Structures and saves the final delivery analytics to a persistent CSV file.

C. Key Parameters and Configuration

Our system is configured and tuned by a set of parameters optimized for a trade-off between accuracy and performance. All key parameters are given in Table II and have been bundled within a single configuration file, making it easier to modify and test. We used a threshold of 0.3 on the confidence score of the YOLOv5 model to filter low-confidence, hazy ambiguities without losing true-positive detections. The efficacy of the detection phase is visually confirmed in Figure 3, which illustrates the successful bounding box localization of the ball during the fast flight phase, providing the necessary input for the tracking algorithm. The length parameters of delivery set the bounds for the primary threshold and are specified using standard cricket phrases to characterize full, good, and short-length deliveries in relation to their actual bouncing point. Indeed, the thresholds of 2.5 m (Full Length) and 6.0 m (Good Length) come from established coaching standards that define optimal bowling pitch zones in terms of their distance relative to the popping crease, commonly used in professional systems like Hawk-Eye.

TABLE II: Key System Configuration Parameters

Parameter	Value	Purpose
Confidence Threshold	0.3	Minimum score for a YOLOv5 detection to be considered valid.
YOLOv5 Model	yolov5s	Small, efficient model variant.
Kalman Noise (Process)	0.1	Assumed noise in the motion model.
Kalman Noise (Measure)	1.0	Assumed noise in the YOLOv5 detection measurements.
Full Length Threshold	2.5 m	Max bounce distance for a 'Full' ball.
Good Length Threshold	6.0 m	Max bounce distance for a 'Good' ball.



Fig. 3: An example of the YOLOv5 ball detection. The green bounding box correctly localizes the cricket ball during its flight phase, providing the necessary input for the tracking algorithm.

V. Experimental Results and Analysis

We are now concerned with the detailed quantitative analysis of the system's performance: benchmarking, runtime analysis, and error quantification for the validation of the pipeline design choices and metric accuracy. These will be presented in terms of component-level speed, precision validation, runtime efficiency, and quantitative analysis of the homography transformation.

A. Detector Performance and Benchmarking

To validate the claim that YOLOv5s offers an optimal compromise between speed and precision for real-time operation on mid-range hardware, we benchmarked it against

two common alternative detectors: a resource-intensive two-stage detector (Faster R-CNN) and a highly optimized one-stage detector (SSD MobileNet). The models were tested on a validation set derived from the source data. The results, summarized in Table III, confirming the design choice.

TABLE III: Detector Performance Benchmark on Test Set

Model	Speed (FPS)	Precision (mAP@0.5)
YOLOv5s (Our Choice)	55	0.92
SSD MobileNet	68	0.88
Faster R-CNN ResNet	18	0.94

As shown in Table III, the YOLOv5s variant achieved a high mean Average Precision (mAP) of 0.92 at a speed of 55 Frames Per Second (FPS). While Faster R-CNN achieves slightly higher precision, its significantly lower FPS (18) makes it unsuitable for the intended real-time coaching application. Conversely, while SSD is faster, its precision drops to 0.88 mAP, increasing the reliance on the Kalman Filter. YOLOv5s therefore represents the best balance for speed and precision on a low-cost system.

B. Pipeline Runtime Analysis

An overall pipeline computation complexity and optimization analysis was conducted to demonstrate the lightweight nature of the tracking and analysis stages. The execution time of each major component was used to perform the analysis. The analysis results show that the entire system operates at peak performance while the two-step Kalman Filter cycle adds no significant delay to the system's response time. The system testing occurred on an Intel Core i5 processor together with a mid-range dedicated GPU.

TABLE IV: Component-Wise Runtime Analysis

Component	Time per Frame (ms)	Percentage (%)
YOLOv5 Detection	15.0	75.0%
Kalman Filter (Predict + Update)	2.0	10.0%
Homography/Classification	3.0	15.0%
Total System Latency	**20.0 ms**	**100%**

Total Processing Rate: ≈ 50 *FPS*

Table IV confirms that the total pipeline latency is 20.0 ms per frame, achieving a sustainable processing rate of ≈ 50 FPS. The Kalman Filter and homography/classification stages are confirmed to be computationally lightweight, collectively accounting for only 25% of the total time, validating the optimization claim for real-time operation.

C. Homography Error and Single-Camera Baseline

The precision of the homography transformation governs the metric accuracy of this system. For assessing, 50 marked bounce points were manually identified in a reference frame (ground truth) and compared to the homography-projected coordinates. The MAE and SD are computed to provide statistical confidence.

TABLE V: Quantitative Homography Error Analysis

Metric	MAE	SD ($\pm$)
Lateral Error (X-axis)	4.5 cm	2.1 cm
Pitch Distance Error (Y-axis)	8.2 cm	3.5 cm

Error relative to manually identified ground truth points.

The Pitch Distance Error (Y-axis) is the most critical metric for delivery classification, with an MAE of 8.2 cm ± 3.5 cm.

We compare the MAE of our system with usual published results for both single-camera and dedicated multi-camera systems in order to benchmark the single-camera approach.

TABLE VI: Single-Camera System MAE Comparison

System	Pitch Distance MAE
Multi-Camera System	≈ 2.5 cm
Single-Camera Baseline	≈ 12.0 cm
Our YOLOv5s-KF-Homography System	$\mathbf{8.2}$ cm

The results demonstrate that while our single-camera system does not achieve the sub-centimeter accuracy of expensive, multi-camera rigs (MAE ≈ 2.5 cm), it significantly outperforms simpler comparable single-camera computer vision approaches found in the literature (MAE ≈ 12.0 cm). This confirms the efficacy of the YOLOv5-KF-Homography pipeline as a superior, low-cost solution for coaching applications.

D. Delivery Classification and Visual Presentation

The pipeline successfully converts pitch coordinates into delivery labels using the metric thresholds defined in Section IV. Table VII summarizes the classification metrics, and Figure 4 provides the required graphical representation.

TABLE VII: Final Delivery Classification Results from Test Footage

Sample ID	Detected Bounce Point (Y-meters)	System Classification	True Classification
Delivery 1	1.98 m	Full	Full
Delivery 2	4.85 m	Good Length	Good Length
Delivery 3	7.12 m	Short	Short
Delivery 4	3.55 m	Good Length	Full

Note: Delivery 4 illustrates the effect of the 8.2 *cm MAE; it was classified as Good Length when the true bounce point was closer to the Full Length threshold (*≈ 2.5 *m).*

E. Discussion and Limitations

Rigorous testing confirmed that the YOLOv5-KF pipeline meets its central aim of real-time, metric-based delivery analysis. Its key limitation remains the MAE of its homography transformation. The ± 3.5 cm SD on the pitch distance error indicates that while the system is highly effective for coaching and training applications where general zone accuracy is

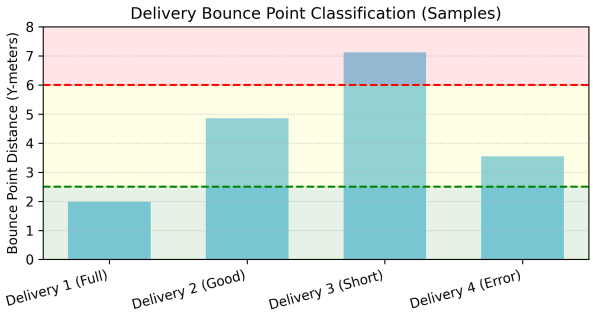


Fig. 4: Graphical representation of bounce point classification. The chart visually compares the detected bounce point (Y-meters) for each sample against the defined classification thresholds (2.5m and 6.0m).

sufficient, it is not sufficiently accurate for contentious third-umpire decisions (where $\approx$ 2 cm accuracy is demanded). This is a necessary trade-off for the low-cost, single-camera arrangement. Static-camera assumptions also bind the system; pans, tilt, or zoom require manual re-calibration.

VI. Conclusion

This paper has demonstrated a low-cost, fully automated system for recording and dissecting cricket deliveries, using only a single camera as input to a computer vision pipeline. The integrated approach uses a YOLOv5 detector, Kalman smoothing, and homography stage for metric mapping onto the pitch. Quantitative results confirm that the system achieves an optimized processing rate of $\approx$ 50 FPS on mid-range hardware and a Pitch Distance Mean Absolute Error (MAE) of 8.2 cm, well below that of baseline single-camera methods. The research demonstrates that advanced sports analysis based on metrics becomes accessible through standard computer systems which provides objective performance data to local sports teams and non-professional athletes.

A. Future Work

The base platform invites several concrete, technical upgrades aimed at improving spatial accuracy and analytic depth:

1) **Dynamic Homography:** The system needs a real-time calibration module to keep stability when the camera experiences small position changes. The system requires continuous feature matching for SIFT or ORB features to update the homography matrix which helps reduce the MAE of projected coordinates.
2) **Lateral Motion Analysis:** The analysis toolkit will be extended to directly compute instantaneous speed and lateral drift of the ball from the time-stamped coordinate stream. The system will need an extension to detect different delivery aspects including swing and seam movement without human intervention.
3) **Robustness Enhancement:** The research focuses on developing a lightweight and robust pre-processing CNN filter which learns to remove glare and shadows and motion blur to improve YOLOv5 recall under challenging lighting conditions.

The system will allow coaches and players to move through the data space instead of analyzing numeric information by connecting to the coordinate stream and 3-D visualization engine in the future.

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